Transferring and Reading Skills in Language Model Contexts

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ABSTRACT

When a language model benefits from a procedure supplied in context, can its internal representation replace the source document, and when does the model cease to depend on that source? We study this question by pairing two interventions: transferring states from a correct-skill run into a matched wrong-skill run, and blocking subsequent attention to the skill in the original correct-skill run. Their depth profiles reveal an intermediate regime in which skill states have lost most of their transfer effect but attention to those states remains important for successful answers. On Qwen3-8B with clinical calculators, the two half-effect boundaries occur at layers 13 and 25 of 36. The ordering holds across three models and two task domains. To establish what is being transferred, we isolate content-specific state differences and test their position, question dependence and rank. Early replacement across the skill span recovers 0.89 of the gain on a screened 135-item subset, whereas a final-position patch is strongly answer-format dependent. Recovery requires appropriate content, placement and multiple leading directions. Repeated late replacement fails to close the depth gap, and an apparent spectral collapse is explained by unusually large activations. These results separate transferability from continued source dependence and provide behavioural criteria for evaluating context representations and compression.

CCS CONCEPTS

• **Computing methodologies** → **Natural language processing**; *Knowledge representation and reasoning*.

KEYWORDS

large language models, skill documents, in-context learning, activation patching, representation rank, mechanistic interpretability

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1 INTRODUCTION

Web search increasingly supplies documents directly to language models, which use retrieved content to answer questions and carry out tasks. This makes document delivery part of the computation: systems must decide what content to retain and what can be replaced by a shorter representation. Work on Web retrieval-augmented generation already shows that the representation of retrieved material affects its usefulness [34]. Procedural documents pose a particular challenge because formulas, conditions and ordered steps must support computation beyond retrieving a fact. *When can the model's internal representation stand in for a supplied procedure, and when does successful computation still require access to its source tokens?* This is the reader-side question we study, motivated by the use of external procedures in retrieval-augmented Web systems. We isolate it with controlled procedural tasks and known-correct documents, holding retrieval fixed.

We call a reusable procedure, formula, lookup table or worked example supplied in context a *skill*. Skills can substantially improve performance [19, 32], but the improvement does not reveal whether the model has formed a reusable representation or still relies on reading the document. Compact states can transfer tasks and instructions [3, 6, 15, 36], with effectiveness depending on position and task scope [20]. A failed transfer, however, has several possible causes: the probe may miss useful states, the receiving computation may be unable to use them, or the source may no longer matter. Distinguishing these possibilities is essential to interpreting what a context representation does.

Our central finding is a separation between *transferability* and *continued reading dependence*. We transplant the skill's states into a run with an unrelated, equal-length skill, then separately block attention to the skill in a correct-skill run. On Qwen3-8B and MedCalc, transfer falls below half of the available gain at layer 13 of 36, whereas blocking subsequent reading still leaves fewer than half of the rescued items correct until layer 25. There is therefore a substantial intermediate depth interval in which the states are difficult to transfer yet remain important to their original computation. This ordering holds in all six evaluated combinations of three models and two task domains. It rules out interpreting loss of transfer as the point at which the model has finished using the source.

We establish an effective transfer probe before interpreting its failure. Matched content subtraction separates the useful procedure from the effect of adding a document. Synthetic answer-format comparisons expose the limits of a final-position probe. Early span replacement then recovers 0.89 of the gain on a screened clinical-calculator subset. Content, cross-question, position and rank controls identify what supports this recovery. Repeated late replacement and spectral diagnostics constrain explanations of its loss.

For context compression and reuse, successful state transfer and safe removal of source access become separate evaluation targets. We contribute:

(1) **A paired test of transfer and reading.** Span replacement and attention knockout reveal an intermediate regime of weak transfer and strong source dependence, replicated across models and task domains (§4.7–4.8).
(2) **A controlled account of the transferred content.** Matched subtraction and answer-format tests motivate

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span interventions. Content, position and cross-question controls identify what makes early transfer effective (§4.2–4.5).

(3) **Constraints on representation compactness.** Rank truncation, repeated late injection and spectral diagnostics distinguish behaviourally useful structure from geometric summaries (§4.6–4.7).

2 RELATED WORK

Transfer interventions and source dependence. Task and function vectors transfer behaviours from demonstrations [15, 36]. Natural-language instructions can yield different function-vector components [6], and instruction-vector patches can interact superadditively across layers [3]. Transfer also depends on time and semantic scope [20]. These results motivate testing continued source dependence separately: Sia et al. [30] use cumulative context masking, and Pola and Balasubramanian [27] compare instruction-span and final-position patches. We pair span transfer with cumulative attention knockout under matched task and model conditions.

What a transferred state measures. A multiple-choice answer symbol [38] and states from completed reasoning [23] differ from the procedural context supplied before generation. Sequential patching [8] and repeated activation steering [21, 26, 31] also intervene beyond our one-time prefill protocol. Following activation-patching methodology [14, 40], we use matched controls, generated answers and identity checks to delimit what recovery establishes. Our depth comparison extends beyond tracing factual recall [11, 24] to the sustained use of procedures.

Skills and compact context representations. Skill benchmarks evaluate retrieval and application [12, 19, 32]. We hold document selection fixed and study the reader. Learned compression uses aggregated activations [1], summary tokens [25], compact states [10] or weight updates [4, 5]. We intervene on naturally occurring states without training a compressor. Dong et al. [7] study single-vector limitations in linear attention and test multiple task vectors. Their task-mapping rank differs from our rank of state differences across procedural positions.

Rank and spectral diagnostics. Spectral studies describe hidden-state geometry [9, 37] and perturbation sensitivity [28]. We instead truncate the injected matrix and measure recovered behaviour. Unusually large activations [29, 33] and dominant dimensions [18, 35] motivate a separate check of whether an apparent spectral collapse actually reflects useful-content compression.

3 SEPARATING TRANSFER FROM READING

Two intervention questions. State transfer tests recovery in a wrong-skill receiver. Attention knockout tests source dependence in the correct-skill run. Both score generated answers while varying intervention depth.

Isolating content before transferring it. Let x be a question, s its correct skill, and $\tilde{s}$ an unrelated skill with the same token length. For a fixed autoregressive model of hidden width D, let $h^L_{\text{skill}}(p)$, $h^L_{\text{wrong}}(p)$ and $h^L_{\text{no}}(p)$ denote residual-stream hidden states at layer L and prompt position p with s, with $\tilde{s}$ and without a skill. At the final prompt position, we decompose the total displacement as

$$\underbrace{h_{\text{skill}} - h_{\text{no}}}_{t\ \text{(total)}} = \underbrace{h_{\text{wrong}} - h_{\text{no}}}_{g\ \text{(presence)}} + \underbrace{h_{\text{skill}} - h_{\text{wrong}}}_{d\ \text{(content)}}. \tag{1}$$

We omit layer and position indices in Eq. 1 for readability. "Presence" denotes the change induced by the chosen wrong skill. "Content" denotes the additional change induced by replacing it with the correct skill. This algebraic decomposition does not assume independent underlying factors. We test each component by adding it to the corresponding state in a no-skill run.

Across the skill's m token positions S, the content-specific difference is a matrix $d^L_S = H^L_{\text{skill}}(S) - H^L_{\text{wrong}}(S) \in \mathbb{R}^{m\times D}$. We use a run containing the wrong skill as the *receiver*, and the correct-skill run as the *donor*. Injection at dose α sets

$$H^L_{\text{patched}}(S) = H^L_{\text{wrong}}(S) + \alpha d^L_S. \tag{2}$$

At $\alpha = 1$, the donor's states replace the receiver's across the span. Other states remain unchanged at that layer. Computation then continues. Matching lengths preserves positions and their encodings. Span interventions change many more states than final-position interventions, which limits direct comparisons. Here the final position is the prompt's end, after the query. State patches are applied during prefill, after which decoding proceeds without further patches.

Behavioural recovery and evaluation subsets. We score generated answers and measure the fraction of the donor's accuracy gain recovered by an intervention:

$$\rho(\text{arm}) = \frac{\text{acc(arm)} - \text{acc(receiver)}}{\text{acc(donor)} - \text{acc(receiver)}}. \tag{3}$$

Here, an arm is an intervention condition, and the donor baseline uses the correct skill in context. Thus $\rho = 1$ recovers the donor's observed gain, and values above one indicate greater accuracy than the donor. We distinguish rescued items (R: wrong without the skill, correct with it), persistent failures (F: wrong under both), kept items (K: correct under both), and broken items (B: correct without, wrong with). Main causal comparisons use rescued items. For span interventions, we further retain only skill groups for which the wrong-skill receiver solves none of the sampled items. We apply this screen separately within each run, model and task, without using intervention outcomes. The estimates are conditional on baseline failure in the sampled group, rather than recovery over the full benchmark.

Measuring continued reading. Cumulative attention knockout masks the skill positions as attention keys at every layer from L onward, during both prefill and decoding. On the knockout run's rescued items within the screened groups, let $q(L)$ be the fraction still solved after this intervention. A low $q(L)$ means that removing subsequent access is damaging. The transfer boundary is the first layer with $\rho(L) < 0.5$ after early successful transfer. The reading boundary is the first with $q(L) \geq 0.5$. The interval between them has weak transfer yet strong reading dependence under these thresholds. These are summaries of two curves with different intervention backgrounds, not a common measure of information content.

Testing the structure of transferable states. To test how many directions support recovery, we subtract the mean across positions, truncate the resulting matrix, and restore the mean:

$$d^{(k)} = \mathbf{1}\bar{d} + \sum_{i=1}^{k} \sigma_i u_i v_i^\top, \qquad d - \mathbf{1}\bar{d} = \sum_{i} \sigma_i u_i v_i^\top. \tag{4}$$

Here, $\bar{d} \in \mathbb{R}^{1\times D}$ is the mean state difference, repeated at every position. The singular values σ_i are ordered from largest to smallest, with u_i and v_i their associated directions. Thus k counts retained directions after centring, and the complete injected matrix has rank at most k+1. A bottom-k control keeps the k smallest singular directions and the same mean. We also vary the intervention layer. Span replacement tests whether states at that layer suffice to transfer the effect. Blocking attention to the span from that layer onward tests whether subsequent computation still depends on reading it.

4 EVIDENCE FOR TRANSFER WITHOUT SOURCE INDEPENDENCE

4.1 Experimental setup

Tasks and models. Our main evaluation pairs MedCalc-Bench [17] with skills from SRA-Bench [32], covering 55 clinical calculators with 20 questions each (1,100 items). Qwen3-8B [39] generates step-by-step reasoning using greedy decoding, without tools. We evaluate its answers with the benchmark scorer. Accuracy is 74.2% with the correct skill, 34.5% without a skill, and 31.5% with a wrong skill. For the decomposition and answer-format experiments, we use a synthetic unit-conversion task with a controlled lookup table and matched wrong skills. We also test Qwen3-0.6B, Qwen3-1.7B and Mistral-7B-Instruct-v0.3 [16], and evaluate transfer to SRA-Bench's TheoremQA subset (747 questions and 320 skills) using its prompt and scorer.

Sampling and uncertainty. The main MedCalc span experiments cover 467 rescued items from 48 calculators with at least two such items. Screening for groups whose receiver solves no sampled item retains 135 items from 14 calculators. The main depth sweep, rank curve and span-quarter comparisons use this same subset. TheoremQA uses the same screen, becoming item-level when a skill has only one question. Intervals resample skill groups, with donor and receiver baselines matched to each intervention's available items. Only complete reruns enter curves. Appendix B records earlier development samples.

Identity control. Writing the receiver's own states back should leave its output unchanged. This identity control exposed a mismatch: capture omitted an attention mask that decoding supplied, changing 108/1,766 Qwen3-8B outcomes (6.1%). Matching the masks restores exact agreement in every corrected run, for example on all 467 MedCalc items and on all 9,308 item–layer pairs of the synthetic decomposition. All main-text interventions use the correction. Appendix A.2 records provenance and Appendix A.1 implementation details. Small numerical discrepancies can propagate through long generations [13].

Table 1: Final-position state replacement across answer formats, Qwen3-8B. All formats use the same 358 items and the same skill, with every layer (0–35) tested. Recovery is the best state-replacement result over layers on each format's rescued items. The identity intervention reproduces the unpatched output on every item and layer.

Answer format	Tokens	Evaluated	Rescued	Best recovery
Multiple choice	1	358	94	1.000
Numeric answer	1–5	358	119	0.235
Chain of thought	Hundreds	358	354	0.003

4.2 Establishing a content-sensitive transfer probe

Content gain and state displacement need not agree. Across the 1,100 clinical-calculator instances, the correct skill adds 39.7 accuracy points over no skill, while the matched wrong skill changes accuracy by −2.9 points (95% interval $[-8.6, +2.9]$, paired calculator bootstrap). Yet on the synthetic task, correct and wrong skills have similar final-position displacements relative to no skill: cosine similarity is 0.91–0.95 over layers 0–10. A large shared displacement is therefore not sufficient evidence that the probe captures useful content. We test the two components separately rather than treating the total displacement as a skill.

Content subtraction recovers the behavioural effect. On the synthetic task with Qwen3-1.7B (358 items, 82 rescued), we inject each component of Eq. 1 at the final prompt position. On rescued items, the presence component recovers 0.098 of the gain, exactly the wrong-skill baseline. The content component recovers 0.805, compared with 1.000 for full state replacement. Recovery also depends on the subtraction baseline. Using a skill with corrupted numbers or with shuffled sentences gives 0.17–0.28 over layers 21–27, every layer measured, compared with 0.805 for an unrelated skill. The content component must therefore be interpreted relative to the specified wrong skill.

Single-position recovery depends on answer format. Table 1 compares answer formats on the synthetic task. All three formats use the same 358 items and all 36 layers. For short numeric answers, no final-position intervention recovers more than one of the 119 rescued items through layer 22. The best results across layers are 0.235 for state replacement, 0.21 for content and 0.008 for presence. These answers are one to five tokens long (341 of 358 use two to four). Patching the last 1, 4, 16, 32, 48 and 61 positions (61 covers the whole no-document prompt) recovers 0.235, 0.261, 0.403, 0.403, 0.395 and 0.370: the ladder saturates at 16, and giving the patch every position the prompt has does not raise it. For step-by-step reasoning, the skill raises accuracy from 0.000 to 0.989, yet single-position replacement recovers at most one of 354 rescued items (0.003) at any layer. Recovery varies with answer format, which changes both output length and computation. Same-query replacement of the final-layer final state also copies the next-token decision. We therefore interpret the comparison as a test of this one-time intervention, alongside prior evidence that transfer depends on position and task scope [20].

The corrected clinical comparison tests the final position and larger sets of positions after the skill at layer 8, using the same 135 items and wrong-skill receiver as span injection (Figure 1). The last 1, 16 and up to 256 positions recover 0.03, 0.06 and 0.05, respectively. Earlier multi-layer probes are reported separately in the extended version. The complete matched comparison here concerns layer 8.

Readout checks. Generated-answer accuracy is the primary readout. Gold-answer log-probability can favour an intervention with worse accuracy, and forcing an answer cue can suppress the skill's benefit. Multiple-choice controls also retain substantial performance after content corruption [2] (extended version).

4.3 Recovering the effect at its source positions

Changing the intervention positions restores recovery. We next apply Eq. 2 across the skill's own positions. At layer 8, span injection raises accuracy from 0.000 to 0.815 on the screened 135-item subset, recovering 0.89 of the donor's gain (Table 2). Figure 1 compares this with writing donor states at the final position, the final 16 positions, or up to 256 positions after the skill, using the same layer, items and receiver. All positions outside the chosen intervention retain their receiver states.

Dose, transfer, order and noise controls. Recovery changes nonlinearly with injection strength. On the screened subset ($n = 135$, 14 calculators), $\alpha = 0.4, 0.5, 0.55$ gives $0.04, 0.04, 0.14$ recovery, while $0.6, 0.7, 1.0$ gives $0.53, 0.87, 0.89$. Doubling the injection to $\alpha = 2$ reduces recovery to 0.77. Most of the increase occurs between 0.55 and 0.7. To test content specificity, we compare correct and other-skill injections over a common prefix length: the correct prefix recovers 0.65, whereas content from an unrelated skill recovers 0.05. A same-family mean control recovers 0.12 on its 41 eligible items. Permuting positions also gives 0.05, norm-matched Gaussian noise gives 0.02, and the identity control gives 0.00. These controls show that recovery depends on the injected content and its arrangement across positions, rather than perturbation size alone.

These results are conditional on the receiver screen. On excluded groups, other-skill and noise controls perform better, limiting extrapolation to the full benchmark. Span patches also write many more states than a final-position probe. We address this size difference next.

4.4 Distinguishing position budget from placement

Span injection writes 365–1,630 positions, whereas the final-position probe writes one. We compare position selections at fixed written-token budgets B, using the same 135 items, layer and receiver as Table 2 (Figure 2). *Quantity alone is insufficient.* Evenly spaced or random subsets recover at most 0.15, even at half the span. Choosing the positions with the largest content change recovers at most 0.12. Writing after the span stays at or below 0.07, at layer 8 or the seven tested layers from 4 to 30. *Block selections differ.* Selecting 256 positions as one block recovers 0.37. Selecting them as 2, 4, 8, 16 and 32 spaced blocks gives 0.24, 0.22, 0.13, 0.08 and 0.08. The one-versus-32 difference is 0.29 $[0.09, 0.48]$, with calculator-bootstrap uncertainty. These selections have similar mean energy shares (0.291 versus 0.295), but energy is not matched per item and the selected content changes: procedure coverage is 38.6% versus 30.7%. This does not isolate contiguity from semantic coverage. *Section and placement matter.* A half-span block centred on the procedure recovers 0.82 $[0.67, 0.93]$, versus 0.18 around the example. Shifting each content difference by one position gives 0.15–0.22 with or without wrap-around. Omitting the first or last 16 positions but leaving the rest in place preserves 0.89–0.92. Recovery is thus sensitive to which content is injected and where, beyond the number of written positions.

4.5 Testing whether transfer carries reusable content

A successful same-question patch could carry a task-specific answer rather than reusable procedural content. Table 3 tests this alternative by varying the injected section and the donor question. Of four contiguous span quarters, the formula quarter gives the largest standalone recovery (0.56), compared with 0.89 for the full span. With the skill before the question, causal masking prevents its states from depending on later question tokens: states are identical across instances at equal prompt length and nearly identical otherwise (cosine ≥ 0.994 over all 48 calculators). Placing the skill after the question allows such dependence. Even then, states from other instances of the same skill recover 0.98 at layer 8, close to the current instance's 1.02. At layer 16, the gap widens to 0.29 versus 0.49, while the instance-specific remainder recovers little on its own. Early transfer therefore relies largely on content shared across questions, with greater question dependence later.

Two completed corrected controls test whether a shared shift or a change in state norm could explain recovery. Averaging content differences across skills recovers only 0.03 with the skill first ($n = 135$) and 0.04 with it last ($n = 151$). Rescaling patched states to match the receiver's norm at each position preserves 0.87 recovery, compared with 0.89 without rescaling. In the corrected rerun, layer-16 recovery is 0.49 with the skill last (151 items, 17 calculators) and 0.06 with it first (135 items, 14 calculators). Because these screens retain different subsets, the contrast suggests sensitivity to prompt order but is not a matched estimate of its effect (extended version).

4.6 Testing a compact representation of the span

We next inject the truncated matrix in Eq. 4 to measure how recovery depends on the retained directions. Each intervention preserves the same skill-specific mean at every span position, so the comparison tests how much variation across positions is needed beyond that mean. At layer 8 on the screened MedCalc subset (135 items, 14 calculators), recovery is $0.07, 0.05, 0.12$ for $k = 4, 16, 32$, rising to 0.63 at 64 and 0.77 at 128, and reaching the untruncated 0.89 at 256 (Figure 3). Keeping the bottom 4 to 128 directions recovers at most 0.04. Recovery therefore depends on which directions are retained: the leading directions support most of the measured effect, but a small number is insufficient under this intervention.

For comparison across models, we summarise each curve by k^*: the first rank reaching halfway from recovery at $k = 16$ to

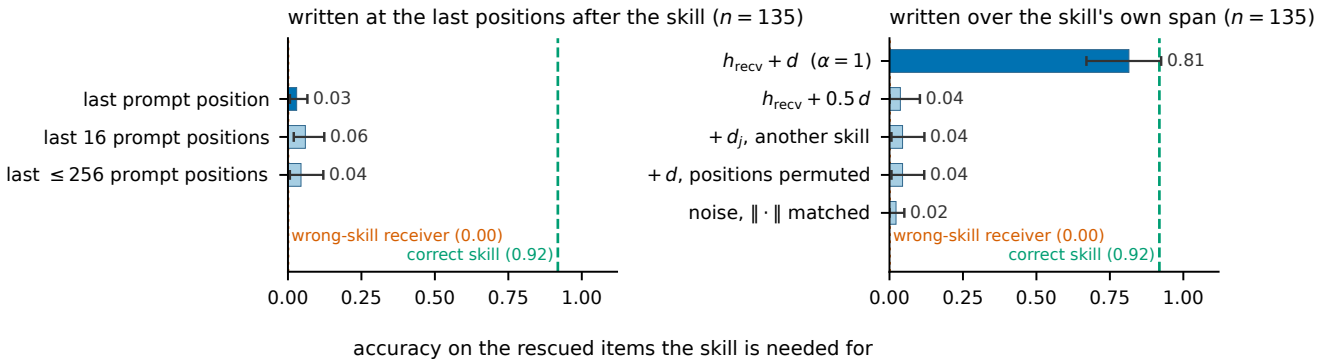


Figure 1: Matched position comparison on MedCalc with Qwen3-8B after the fix. Both panels use layer 8, the same wrong-skill receiver and 135 items from 14 calculators. Left: donor states written at the last 1, 16, or up to 256 available positions after the skill. Right: interventions over the skill span. The dashed line marks correct-skill accuracy; the dotted line marks receiver accuracy. Error bars are calculator-bootstrap 95% intervals.

Table 2: Span interventions on the screened MedCalc subset: 135 items from 14 calculators, selected from 467 rescued items across 48 calculators by requiring the wrong-skill receiver to solve no sampled item in a group. The same-family control is available on 41 items. Each row uses baselines matched to its own items in Eq. 3; accuracy intervals resample calculators. Table 4 gives the interval for full-span recovery.

intervention	n	accuracy	CI_{95}	recovery ρ
correct skill in context	135	0.919	[0.844, 0.976]	+1.00
$h_{recv} + d$ (full span)	135	0.815	[0.670, 0.925]	+0.89
correct content, shared prefix	135	0.593	[0.336, 0.820]	+0.65
half dose ($\alpha = 0.5$)	135	0.037	[0.000, 0.103]	+0.04
another skill (individual donor)	135	0.044	[0.007, 0.119]	+0.05
mean from another skill family	135	0.044	[0.012, 0.102]	+0.05
mean from the same skill family	41	0.122	[0.000, 0.250]	+0.12
positions permuted	135	0.044	[0.007, 0.118]	+0.05
norm-matched Gaussian noise	135	0.022	[0.000, 0.051]	+0.02
identity intervention	135	0.000	[0.000, 0.000]	+0.00
wrong skill, unpatched	135	0.000	[0.000, 0.000]	+0.00

the maximum on the fixed grid $\{16, 32, 64, 128\}$, with interpolation in $\log_2 k$. We call this the *rank knee*, a summary of the measured recovery range. Qwen3-8B gives $k^* = 47$ [43, 55], approximately 1.1% of its hidden width. The extended version reports development and deeper-layer curves. These measurements describe what the injected representation can recover, rather than directly measuring the model's unmodified computation: truncated states may activate pathways absent from an unpatched run [22].

4.7 Identifying the interval between transfer and reading

The preceding controls establish effective early transfer of content across skill positions. We now ask whether losing that effect at later layers means the model no longer needs the document. Figure 4 tests this inference directly.

Transfer weakens while reading remains important. Figure 4 compares transfer through span replacement with continued use through attention on Qwen3-8B. Every layer is measured, and both interventions are read on the same screened items. Replacement recovers 0.92–0.98 through layer 7, 0.89 at 8 and 0.74 at 12 of 36, then 0.16 at 13 and at most 0.07 from 14 onward ($n = 135$, 14 calculators). The drop between layers 12 and 13 accounts for most of the decline. We define the crossing below $\rho = 0.5$ as the *transfer boundary* (labelled "handover" in the figures and tables). The log-probability of the donor's generated reasoning also drops between these layers, suggesting that binary answer grading sharpens, but does not fully explain, the transition (Appendix B). To test continued reading, *cumulative attention knockout* blocks attention to skill tokens at every layer from L onward. On the 134 rescued items of the same subset, 0.03–0.12 stay correct for $L \leq 23$. This rises to 0.25 at 24, first passes half at 25 (0.58) and reaches 0.94 at 35. The transfer boundary therefore lies at $13/36 = 0.36$ of depth and the reading boundary at $25/36 = 0.69$, using the same screen and restricting knockout to its donor-correct items. Writing the last prompt position instead recovers at most 0.06 at any layer. Continued dependence on skill attention thus outlasts successful transfer from the skill's states. Transfer measures recovery in a

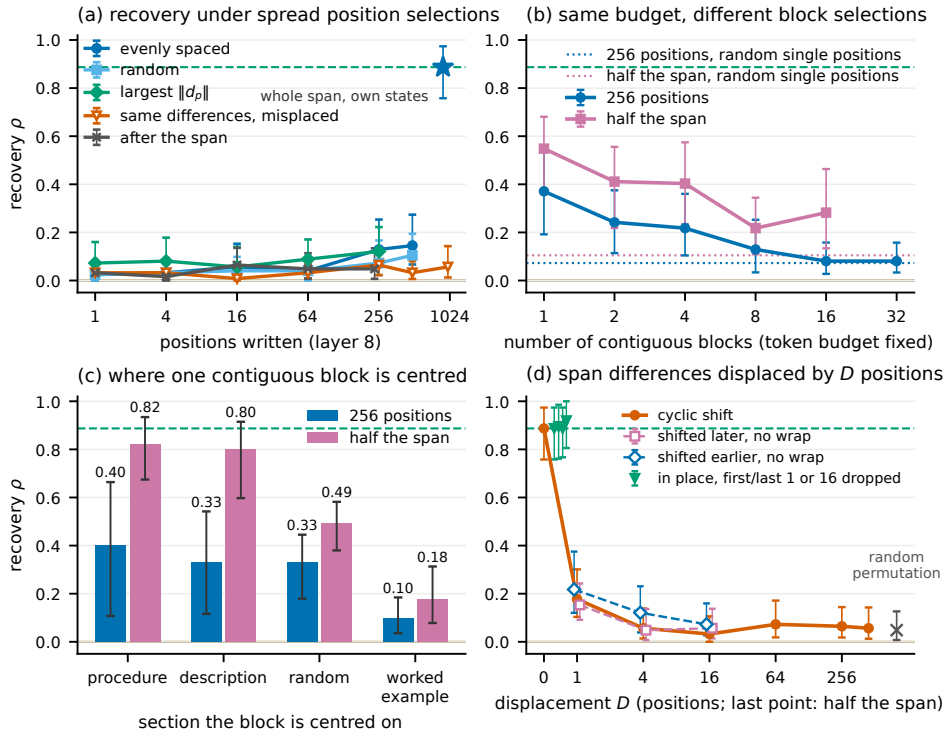


Figure 2: Position budget on MedCalc with Qwen3-8B, layer 8: the same 135 screened items (14 calculators) and wrong-skill receiver in every panel; the dashed line is the whole span written in place ($\rho = 0.89$). (a) B positions, each written with its own donor state (evenly spaced, random, or largest $\|d_p\|$), the corresponding differences written half a span away, or the last B prompt positions after the span. (b) A fixed budget written as N contiguous blocks; dotted lines are random single positions at the same budget. (c) One contiguous block centred on a section of the skill. (d) The span differences displaced by D positions, cyclically or without wrap-around; triangles write it in place without its first or last 1 or 16 positions. Error bars are calculator-bootstrap 95% intervals.

wrong-skill receiver, whereas knockout measures dependence in a correct-skill run. The separation identifies the intermediate regime operationally: from layers 13 through 24, span replacement recovers less than half the gain and blocking further reading leaves fewer than half the rescued items correct. From layer 25, at least half remain correct without further reading, with residual dependence continuing to diminish. Thus transfer loss precedes, rather than demonstrates, tolerance to removal of source access.

Repeated replacement does not bridge the gap. To test whether late failure reflects intervening at too few layers, we replace span states repeatedly within consecutive layer windows, using the battery's receiver and the same 135 items as the depth sweep. Windows 8–11, 12–15, 14–19 and 16–35 recover 0.89, 0.74, 0.16 and 0.06. The last equals layer 16 alone. Writing every layer at once recovers 0.98 [0.91, 1.03], so the transplant itself remains effective when it starts early enough. The identity control is exact on all 467 items. On TheoremQA (99 items, 75 groups, with its battery's receiver), the same windows recover 0.84, 0.61, 0.38, 0.20, the last again equal to layer 16 alone. Repeated late injection therefore does not restore early-layer recovery in these tests. This constrains a simple shortage-of-interventions explanation, while leaving interactions with the receiver's other states open [3].

A spectral drop is not evidence of compression. Could the loss of transfer reflect compression into fewer dimensions? We examine the centred participation ratio, a summary of how broadly matrix energy is distributed across singular directions. It falls from 68 at layer 12 to 1.0 at layer 16, but this drop is driven by unusually large activations [29, 33]. In 35 of 48 calculators, one token's state reaches 121–195 times the median span norm, always on the same two hidden dimensions. Removing the four highest-norm positions restores the ratio to 85 at layer 16, and it stays between 62 and 98 over layers 0–28. The drop is also absent in Qwen3-0.6B and Qwen3-1.7B. This diagnostic therefore does not support compression as the explanation for lost transfer: a few large activations distort the dimensionality summary (extended version).

4.8 Replication across models and task domains

We extend span injection, rank truncation and the available depth comparisons to Qwen3-0.6B ($D = 1024$), Mistral-7B ($D = 4096$,

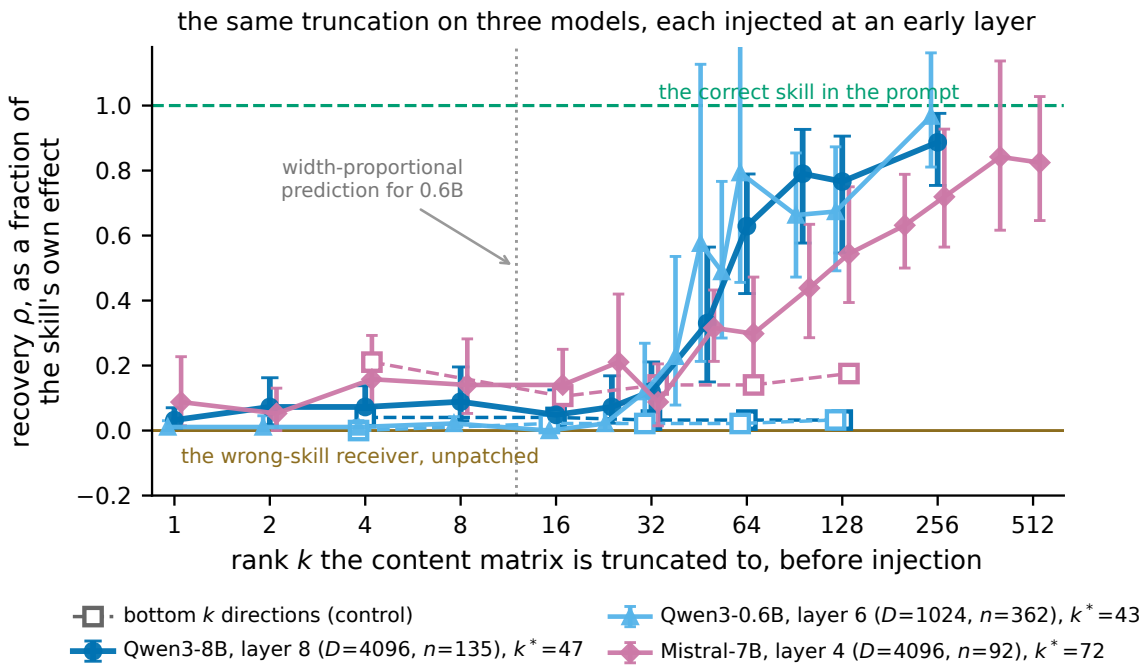


Figure 3: Recovery after retaining k leading centred directions and restoring the positional mean, on MedCalc after the fix. Each model is injected at an early layer. Despite a fourfold difference in hidden width, Qwen3-0.6B and Qwen3-8B have similar rank knees (43 and 47); Mistral-7B, with the same width as Qwen3-8B, has $k^* = 72$ [46, 94]. Untruncated recovery is 0.89, 0.91 and 0.84, respectively, on each model's screened subset (135, 362 and 92 items). Both Qwen models reach that value by $k = 256$ and Mistral-7B at $k = 384$. Dashed curves retain bottom directions. Error bars are calculator-bootstrap 95% intervals. Only complete rank experiments are plotted.

Table 3: Content and question dependence of span states, Qwen3-8B after the fix. Top: each contiguous quarter is injected separately, on 135 items from 14 calculators. Bottom: with the skill after the question, content shared across other instances and the instance-specific remainder are injected separately, on 151 items from 17 calculators. Each comparison uses its screened subset from the larger rerun. All entries report recovery ρ.

Skill quarters, layer 8, skill before question (n = 135)	
Injected content	Recovery ρ
Prose description	0.09
Formulas	0.56
Tool signatures	0.12
Worked example	0.05
Full span	0.89

Question dependence, skill after question (n = 151)		
Injected content	Layer 8	Layer 16
This instance	1.02	0.49
Other instances of the same skill	0.98	0.29
Instance-specific remainder	0.03	0.05

32 layers), and TheoremQA. Each model is injected at an early layer, and the receiver screen is applied separately to each run. Table 4 collects the paired transfer and reading boundaries alongside the early-transfer and rank results. Results for Qwen3-1.7B, where skills rescue only 8–11% of items, appear in the extended version.

Span recovery ranges from 0.74 to 0.94 across the six model–task combinations. The largest control is 0.35, for a same-family mean on 31 Mistral/MedCalc items, where full-span recovery on the same items is 0.71. In every model–task combination, the transfer boundary occurs before the reading boundary. The measured transfer boundaries lie at 0.33–0.43 of model depth within Qwen and at 0.19–0.22 in Mistral. Reading boundaries also vary by task: 0.58–0.71 on TheoremQA and 0.69–0.84 on MedCalc across the three models.

The pre-registered width-proportional prediction places Qwen3-0.6B's rank knee near 12. Instead, it is 43 [39, 52], close to Qwen3-8B's 47 [43, 55] despite the fourfold width difference. TheoremQA gives 52 [46, 62] and 50 [39, 67], respectively. Mistral gives $k^* = 72$ [46, 94] on MedCalc and 64 [50, 79] on TheoremQA. These values are higher than Qwen3-8B's but have overlapping intervals. Thus hidden width alone does not determine the measured rank requirement, while a difference between families is not established. Full curves and controls are retained in Figure 3 and the extended version.

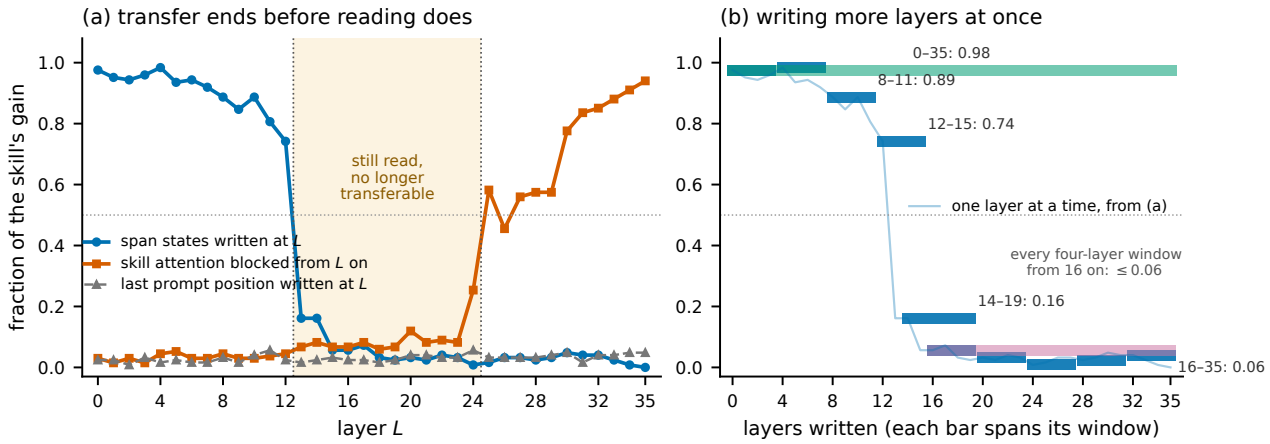


Figure 4: Transfer declines earlier than dependence on skill attention, Qwen3-8B on MedCalc, with every layer measured. (a) Three interventions on one axis and the same screened items (n = 135; the knockout on their 134 rescued items): span states written at layer L; attention to the skill blocked from L onward; and the last prompt position written at L. The shaded band spans the layers where skill attention still supports performance but span replacement recovers less than half of the gain. (b) Each window is drawn across the layers it writes: four-layer windows tile the stack, and the long windows 14–19, 16–35 and 0–35 are shown with them; the faint curve repeats (a). Writing every layer at once recovers 0.98, while any window starting at 16 or later stays at or below 0.06. Span interventions use the corrected implementation; attention knockout does not use state capture.

Table 4: Replication with the corrected implementation. Injection layers are 8 for Qwen3-8B, 6 for Qwen3-0.6B and 4 for Mistral-7B. Recovery intervals resample skill groups. Max control includes half-dose, other-skill, permutation and noise comparisons, including same-family means available on 41 MedCalc/Qwen3-8B and 31 MedCalc/Mistral items. n describes the battery. Handover brackets the transfer boundary as a fraction of model depth; reading is the first layer, at one-layer resolution, from which blocking attention to the skill still leaves at least half of the rescued items correct, measured on the knockout run's rescued items within the same screened groups. k^* is the rank knee on the fixed grid.

task	model	n (groups)	ρ [95% CI]	max control	handover	reading	k^*
MedCalc	Qwen3-8B	135 (14)	0.89 [0.76, 0.97]	0.12	0.33–0.36	0.69	47
TheoremQA	Qwen3-8B	99 (75)	0.84 [0.75, 0.92]	0.25	0.33–0.36	0.58	50
MedCalc	Qwen3-0.6B	362 (38)	0.91 [0.75, 1.05]	0.30	0.36–0.39	0.79	43
TheoremQA	Qwen3-0.6B	132 (97)	0.74 [0.66, 0.82]	0.13	0.36–0.39	0.71	52
MedCalc	Mistral-7B	92 (20)	0.84 [0.69, 1.09]	0.35	0.19–0.22	0.84	72
TheoremQA	Mistral-7B	115 (94)	0.94 [0.84, 1.05]	0.27	0.19–0.22	0.59	64

5 DISCUSSION AND CONCLUSION

What the depth gap establishes. The paired tests distinguish early effective transfer, an intermediate interval of weak transfer with continued reading dependence, and later tolerance to blocking further reading. These operational regimes do not identify separate internal algorithms. The gap is compatible with dependence on the receiving state, whose role remains unresolved.

Implications for representation research. What a state can transfer and whether its source remains necessary are different questions. Content and cross-question controls identify useful procedural information, while position and rank interventions characterise its usable structure. Failed late transfer and distorted spectral summaries cannot alone establish that this information has disappeared or been compressed. Pairing transfer with source-access interventions provides a behavioural way to distinguish these claims.

Implications for context compression and reuse. Representations proposed for reuse should be tested both for recovery in the intended receiver and for tolerance to removing source access. Our findings show why token count, state norm and single-format success are insufficient proxies, and why hidden width alone need not determine a suitable rank budget. These are evaluation principles, not demonstrated efficiency gains. The evidence concerns fixed contexts and screened subsets. Layer depth does not specify agent steps or establish when documents can be dropped during multi-turn execution.

AI USE STATEMENT

Generative AI tools assisted with manuscript restructuring and language editing, literature lookup and reference checks, consistency checks between reported results and local experiment records, and

figure-label editing. The authors are responsible for the final text, scientific claims and experimental artifacts.

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A SETUP, IN FULL

Two task families. Real skills, the main material: MedCalc-Bench instances with the 55 gold calculator skills released with SRA-Bench—55 calculators × 20 instances = 1,100 items, graded deterministically against a numeric tolerance with no model in the loop. Both the skills and the tasks are other people's, which is what makes the effect size credible. Qwen3-8B, single turn, no

tools, greedy. Behavioural baseline and replication intervals resample skill groups because items sharing a skill are dependent; original item-level intervals in individual tables and plots are identified separately in the main text. *A synthetic positive control*: a fictional unit system with a one-page document giving its conversion table, so the no-document floor is guaranteed, contamination is impossible, and we know exactly what content means. Items are generated from the table ($n = 358$), and every item whose answer appears in a worked example is dropped.

Two dependent variables. Accuracy, from a greedily decoded free-form answer graded by the task's own scorer, and the teacher-forced log-probability of the gold answer. They fail differently and disagree often enough that reporting one alone is unsafe (§4.2); accuracy is read first throughout. No item was rewritten into multiple choice to obtain an accuracy channel—The extended version shows why that rewrite is not harmless.

Interpreting recovery. Equation 3 normalises an intervention by donor and receiver accuracy on the same items. It measures the fraction of the available gain, not raw accuracy or a percentage of information transferred. Values can exceed one. Comparisons across protocols must retain their own paired baselines.

The instrument's own noise floor, measured. Every decoding run carries a *null* arm: the receiver's own span states are captured and written back over themselves. With a length-matched receiver this is a bit-for-bit identity, so the null arm must equal the unpatched receiver at every layer. Before the fix of Appendix A.1 it did not, quite. On Qwen3-8B, across 1,766 paired observations from every pre-fix run that carries both arms, the null arm agrees with the receiver on 93.9% of items: 108 flip, 46 from wrong to right and 62 from right to wrong, symmetric within a sign test ($p = 0.15$). The pipeline itself is deterministic—the same configuration run on two different machines agreed on all 360 item-by-arm cells—so this is not run-to-run variation but a reproducible property of the instrument: the capture and decoding passes took different attention kernels, which perturbs the written-back state by 1.2% at layer 8, and the dependent variable is the end of a chain of thought several hundred tokens long, which is long enough for that to change an argmax somewhere and propagate. It scales with the model: the same arm flips 0.6% of items on Qwen3-0.6B (1,696 pairs) and 0.8% on Qwen3-1.7B (1,338), against 6.1% on the 8B. After the fix it is exact (80/80).

Three consequences are used throughout. First, **an effect smaller than about six points is not resolvable on the 8B** in pre-fix runs, whatever a binomial interval says. Second, the floor is not uniform: on the persistent-failure cell it is 16%, between two and three times the average, which is what one expects of items on a decision boundary, so numbers read on that cell are read against a coarser ruler. Third, on the restricted split the pre-fix null arm read 0.034 rather than zero—the split conditions on the receiver failing, so a flip there can only go upward—so pre-fix span numbers are read against 0.034; after the fix it reads exactly 0.000, the floor Table 2's 0.750 is read against. We have rarely seen this control reported in the patching literature; at a single answer token it is small enough to be invisible, and it becomes large only once the dependent variable is a long generation.

Four cells. Every item is labelled by what the document did to it: **R** (rescued: wrong without, right with), **F** (persistent: wrong both ways), **K** (kept: right both ways), **B** (broken: right without, wrong with). **R is the only cell in which the document demonstrably works**, every table here is read by cell, and §4.2 shows the aggregate over all four can be sign-inverted relative to R.

A.1 Experimental detail

Models and numerics. Qwen3-1.7B (28 layers, $d = 2048$) and Qwen3-8B (36 layers, $d = 4096$); §4.8 adds Qwen3-0.6B (28 layers, $d = 1024$) and Mistral-7B-Instruct-v0.3 (32 layers, $d = 4096$), whose chat template has no system role, so the system text is merged into the user turn. Greedy decoding throughout, `enable_thinking` disabled and verified by a self-test on the rendered template. Eager attention wherever an attention mask is manipulated, because sdpa and flash silently ignore a 4D mask; layer-output hooks are implementation-independent, so the decode-heavy runs use sdpa, which is about 2.5× faster on a 2k-token prompt. The 1.7B synthetic runs are fp32: bf16 costs 0.5–2.5 nats on the log-probability of a token deep in the tail, and every log-probability quantity here is a difference of such numbers. The 8B runs are bf16 and the gaps we report from them are 5–8 nats, well outside that error; the accuracy channel is immune to it either way.

Self-tests, and one that failed. Three checks run with the experiments. (i) *Self-transplant*: donor and receiver are the same forward, so the patch must be an exact no-op. On the fp32 synthetic runs the deviation is 0.0 nats at every layer. On the bf16 8B runs it is up to 0.38 nats—bf16 arithmetic over two sequence lengths—while the accuracy channel remains an exact no-op at every layer, which is how we tell the two apart. (ii) *Last-layer patch*: replacing the final layer's output at the patched position makes that position's logits the donor's by construction, so the next token must match exactly. (iii) *Query-side control* for the knockouts.

An error worth recording. The first version of the real-skill scoring code patched at index −1 of a forward run over prompt *and* answer, which is the last token of the answer—after every position whose logits are scored. Every arm returned the untouched baseline to six decimals, which is what made it visible. Relative indices are the wrong default whenever the forward length varies between the capture and the patch.

Control documents. The synthetic controls are generated from the skill document by `make_controls.py`, which checks two things: the corrupted document's numbers must not coincide with the task set's gold answers (7 of 249 do, 2.8%), and the shuffle must not be the identity. All three controls are stored under the same filename in separate directories, because the injector derives the "# Skill: *name*" header from the file stem—a file called `SKILL.zorb-units.corrupted.md` would announce the manipulation in the prompt, and the control would then differ from the treatment in two ways.

Item set. The synthetic set is generated from the conversion table with the value range 2–40, dropping any item whose answer also appears in a worked example of the document ($n = 358$; the sweeps use the first 120 or 200). Distractors are values reachable by

misreading the table—the neighbouring row, the wrong family, the inverted direction—never random numbers, and the key is rotated across A/B/C/D so a position-biased model still scores at chance.

A.2 Evaluation scope and intervention provenance

An accidental replication. A queueing mistake ran one 36-layer synthetic sweep (358 items, two arms, generations up to 900 tokens) twice, on two different GPUs in two different containers. All 36 layer files agree byte for byte. This does not bound the nondeterminism discussed in §4.1—it is one configuration on one pair of devices—but it is the only end-to-end check of it we have, and it came out exact.

Sample size and scope. The battery was developed on $n = 40$ (ten calculators × four rescued instances), which gives intervals of roughly ±0.15; §4.3 reruns it on forty calculators and §4.8 on further models and a second task, and the conclusions are the ones that survived that. We report profiles and not individual layers: the one non-monotonicity in the depth curve (0.250 at layer 18 against 0.475 at 20) has overlapping intervals and we do not interpret it. Among the three models, Qwen3-1.7B is the weakest rung: the document rescues only 8–11% of its items, so its truncation curve is not resolved and is reported for completeness rather than relied on. The width argument rests on Qwen3-0.6B against Qwen3-8B, a factor of four within one family; a second family of the same width needs more directions (extended version). In the pre-fix development runs, four independent shards recomputed the three baselines and agreed to the item (0.950 / 0.025 / 0.175), and the no-op arm sits on the receiver's baseline at all twelve layers to within the 5% numerical floor of §4.1, so the instrument is checked even where the estimate is coarse. The span channel, the depth sweep and the truncation curve have three models and two tasks; the knockouts two models and two tasks; the final-position results two models (Qwen3-1.7B, fp32); the document-order comparison and the within-document localisation one.

Which items the causal claims are about. Two restrictions stack. Causal claims are made only on the rescued cell, the only cell in which the document demonstrably works; on the persistent-failure cell the same arms track the document rather than correctness—the transplant fails exactly where the document fails—which is the right behaviour but is not a substitute for the other cells. And within that cell the span arms are read on the calculators whose *receiver* never solves an item, because on the others the model does not need the document, and before the fix any disruption of the wrong one, noise included, let it fall back on what it already knows ($\rho = 0.14$ for noise there against 0.00 here). After the fix noise no longer does that (0.00 on those calculators), but a donor from another skill still reads 0.47 there against 0.19 on the rest. On forty calculators the restriction keeps fifteen of them and 60 items. On this split, accuracy is 0.750 for the transplant and at most 0.083 for the tested negative controls; the main text distinguishes item-level intervals from group-bootstrap uncertainty. Those forty calculators are themselves the 160 rescued items with the most instances per calculator, out of 467. We therefore reran the battery, the truncation curve, the depth sweep, the dose curve and the within-document quarters over *every* rescued item of every calculator with at least two (467 items, 48 calculators; the restriction then keeps 135 items over 14). Every conclusion holds and no control moves above 0.12; Appendix B gives the comparison arm by arm.

What the rank result does and does not establish. §4.6 truncates the injected content matrix to rank k at a layer where the untruncated version recovers most of the effect, and recovery stays near the control floor at $k \leq 32$ while the bottom-k control stays at the floor. That licenses the forward direction: the effect requires directions of the same order as the span's participation ratio, and within the Qwen family neither number scales with width. It does not license a claim that the number belongs to the document alone (Mistral-7B needs more), nor a converse about depth. Supplying a deep layer with a higher-rank message taken from a shallow one does not restore the handover ($\rho = 0.00$ against 0.06 for the same-layer donor, 0.00 with the norm matched), so we do not claim that what a deep layer lacks is rank.

Diagnostic results for the behavioural readout. We could not find any representational quantity at the final position that distinguishes the items a skill rescues from the ones it does not: the norms and angles agree to three digits, and the one similarity measure that separates them under the gold skill separates them equally under a wrong one. And the document acts through a chain of thought, so every causal measurement decodes several hundred tokens; scoring under a forced `ANSWER:` cue instead leaves almost no behavioural effect (0.178 with the skill, 0.131 without and 0.100 with a wrong one, on 720 items) while retaining a +1.08 nat log-probability difference—so the span channel has no usable log-probability reading, and is reported on accuracy alone.

An instrument fault and result provenance. State capture omitted the attention mask supplied during decoding. Although mathematically equivalent for an unpadded sequence, these calls selected different kernels. Their states differed by 1.2% at layer 8 and 25% at layer 34, and identity patching changed 6.1% of Qwen3-8B answers. Matching the masks restores exact identity agreement. All main-text state interventions use the corrected implementation, including the synthetic and final-position sweeps (§4.1). Attention knockout does not use state capture. Historical development results in the extended appendices retain their original sample sizes and provenance. The corrected full-data rerun in Appendix B supersedes the earlier 60-item battery.

Numerics. Log-probabilities on the synthetic control are fp32, because bf16 costs 0.5–2.5 nats on a token deep in the tail. The 8B numbers are bf16, which appears as self-transplant deviations of 0.1–1.1 nats while the accuracy channel remains an exact no-op; accuracy is read first throughout.

B THE FULL-DATASET RERUN, AND WHAT READS THE SPAN

Every MedCalc × Qwen3-8B result in the main text was developed on the forty calculators with the most rescued items, sampling four each: 160 of 467 rescued items, and 60 items over 15 calculators after the receiver-never-solves restriction. We reran each of them

Table 5: Full-dataset rerun against the subset the main text cites, Qwen3-8B on MedCalc, layer 8. Recovery is Eq. 3; intervals are group bootstrap over calculators. *per-calc* is the main text's restriction, *per-item* the weaker one. Controls reach 0.12 under the group screen and 0.25 under the item screen; each arm uses its own matched baselines.

arm	full, per-calc (n = 135)	subset (n = 60)	full, per-item (n = 369)
transplant `real`	**0.89** [0.76, 0.97]	0.87	0.94
same skill, matched prefix	0.65 [0.37, 0.87]	0.73	0.84
half dose $\alpha = 0.5$	0.04	0.06	0.11
positions permuted	0.05	0.08	0.21
norm-matched noise	0.02	0.10	0.18
other calculator	0.05	—	0.25
other family, mean	0.05	0.06	0.16
same family, mean	0.12 [0.00, 0.25]	0.05	0.24
identity patch	0.00 [0.00, 0.00]	0.00	0.00
rank 128 / 64 / 32 / 16	0.77/0.63/0.12/0.05	0.77/0.54/0.13/0.06	0.88/0.70/0.22/0.18
bottom-64 / bottom-16	0.03/0.04	0.00/0.06	0.11/0.12
quarters $q_0/q_1/q_2/q_3$	0.09/**0.56**/0.12/0.05	0.12/0.56/0.12/0.12	0.18/0.64/0.22/0.19

with every rescued item of every calculator that has at least two (467 items, 48 calculators), changing nothing else. The restriction is stricter on the larger set—a calculator now has to fail on up to twenty items rather than four—so it keeps 135 items over 14 calculators, and we also report the per-item restriction (369 items over 45), which keeps the items whose own receiver fails.

The transplant, the truncation knee between 32 and 64 directions, and the formula quarter's dominance all reproduce. The one arm that moves is the same-family donor, from 0.05 to 0.12, on the 41 items where it is defined.

The depth sweep at one-layer resolution. The subset sweep measured layers 0, 4, 8, 12, 14, 15, 16, 20 and placed the collapse somewhere after 12. At full scale we added 13 and 17–19: recovery is 0.98, 0.98, 0.89, 0.74 at layers 0, 4, 8, 12, then 0.16, 0.16, 0.06, 0.06, 0.07, 0.03, 0.02, 0.03 at 13 through 20 (per-calc; the per-item restriction gives 1.00, 0.98, 0.94, 0.79 then 0.32, 0.34, 0.25, 0.25, 0.23, 0.19, 0.14, 0.17). Layers 13 and 14 are indistinguishable under both restrictions, so the transition is a single layer between 12 and 13 and not the gradual decline a stride-2 sweep suggests.

A continuous dependent variable at the same boundary. Accuracy over a chain of thought is a threshold, and a smooth decline inside the model could present as a step in it. We therefore decode the with-document trajectory once per item and teacher-force that same trajectory under every other condition, reading its mean token log-probability: $q = (\ell_L - \ell_{\text{recv}})/(\ell_{\text{gold}} - \ell_{\text{recv}})$ has the same two anchors as ρ with no decoding threshold anywhere in it (n = 442 items on which the document run is correct, 48 calculators). q reads 0.999, 0.989, 0.960, 0.831 at layers 0, 4, 8, 12, then 0.598 [0.550, 0.648] at 13, 0.595 at 14, 0.340 at 16, 0.201 at 20 and 0.129 at 24. Expressed per layer as a fraction of its range, the step at 12 → 13 is 26.8%, against 3.7% per layer over 8–12 and 0.3% from 13 to 14: the discontinuity is present in a readout that cannot be produced by grading. It is smaller than the behavioural step (61% of range), so the grader sharpens the boundary roughly twofold without creating it, and it is spread evenly across the four quarters of the trajectory (0.20, 0.26, 0.31, 0.21) rather than concentrated at the answer, which is the opposite of what a grading artefact predicts. We also tried restricting q to the tokens after the final answer marker and report that it carries no information here: once the full reasoning is teacher-forced, every condition predicts the final number almost perfectly (median ℓ between −0.001 and −0.005 for all of them, and the gold-minus-receiver denominator has median 0.0014, below 0.01 on 59% of items).

Which heads read the span. We recorded each head's contribution from the span to later positions, $W_O^h \sum_{s \in \text{span}} a_{t,s} v_s$, comparing the document and filler runs (n = 455). Raw magnitudes are dominated by the last layers, but the residual stream grows by more than an order of magnitude with depth; normalised by it, the read-out concentrates in layers 12–21 and peaks at 18. It is not a small circuit: of the 1,152 heads the top 1% carry 10.5% of the read-out and the top 5% carry 31.1%, and no head attends to the span more than uniformly (0.4–1.2×, with the span occupying half the prompt). Blocking the 32 highest-ranked heads' attention to the span—ranked on half the calculators and evaluated on the held-out half—costs 0.39 of the document's effect, against 0.23 for 32 random heads at the same layers (paired difference 0.154, 95% CI [0.061, 0.251], McNemar 39 versus 9 discordant, n = 195 over 23 calculators). The ranking is therefore causal, but it buys only 0.15 over chance and 32 heads remove well under half the effect: the span is read diffusely, and we do not claim a nameable set of reader heads. One further observation constrains what fails at depth. After an injection at layer L, the span's own states from $L + 1$ onward are identical to the document run by construction—the span attends only to a shared prefix and to itself—and the downstream per-head read-out of the span is indeed 0.86–0.99 of the document's at every injection layer we tried, including those at which behaviour is gone. This read-out is compatible with continued access to donor content. A late span patch leaves the receiver's earlier question-state updates unchanged, motivating a direct test of their interaction with the transplanted states.